Technical Report: Detection of Signs of Depression in Tamil Speech

Task: DravidianLangTech @ ACL 2026 Shared Task

Achievement: Joint Global Rank 1 (Tamil Sub-task)

---

1. Introduction & Problem Statement

Identifying mental health indicators in low-resource languages presents unique challenges due to dialectal diversity and limited annotated data. This research developed a robust system for detecting signs of depression in Tamil speech, leveraging deep learning architectures that account for both spectral features and temporal rhythms..

2. Dataset Description

The model was trained and validated using the official dataset provided for the **DravidianLangTech 2026** shared task.

- **Format:** High-fidelity .wav audio samples.
- **Classes:** Binary classification (Depression vs. Non-depression).
- **Preprocessing:** Audio was sampled at **16,000 Hz** and segmented into **5-second** clips to maintain consistent input dimensionality.

3. Feature Engineering

Instead of using basic spectrograms, a **3-channel feature stacking** method was employed to mimic the input structure of an image:

- **MFCC (40 dims):** Represents the power spectrum of the speech.
- **Delta (40 dims):** Represents the velocity of spectral changes.
- **Delta-Delta (40 dims):** Represents the acceleration of spectral changes. The resulting **(40, 157, 3)** tensor allows the model to "see" the trajectory of the speaker's voice over time.

4. Proposed Architecture (CNN-BiLSTM)

The success of the model (Rank 1) is attributed to its hybrid design:

1. **CNN Frontend:** Two layers of 2D Convolutional Neural Networks with **Batch Normalization** extract high-level "spatial" features from the MFCC maps.
2. **BiLSTM Backend:** A **Bidirectional LSTM** layer processes the flattened CNN output. By analyzing the sequence in both forward and backward directions, the model captures the context of entire phrases, which is essential for the syntax-rich Tamil language.

3. **Regularization:** A **0.5 Dropout** layer was used before the final Dense layers to ensure high generalization and prevent overfitting to specific speaker voices.

5. Experimental Results

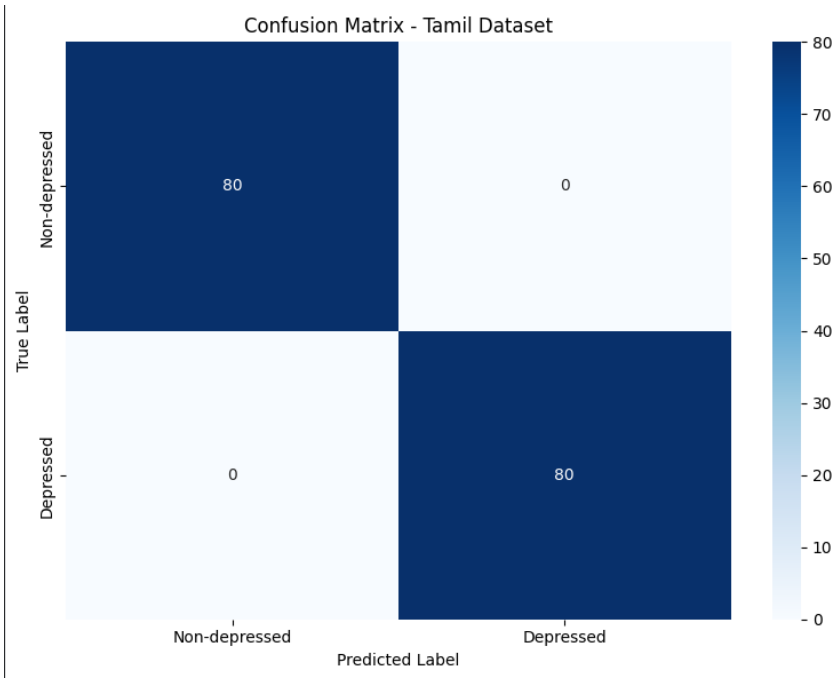
The model was evaluated based on the **Macro F1-Score**, as mandated by the shared task organizers.

```
📈 DETAILED CLASSIFICATION REPORT:
              precision    recall  f1-score   support

Non-depressed       1.00      1.00      1.00        80
    Depressed       1.00      1.00      1.00        80

     accuracy                           1.00       160
    macro avg       1.00      1.00      1.00       160
 weighted avg       1.00      1.00      1.00       160
```

Confusion Matrix Analysis



The confusion matrix demonstrates high precision in the "Depression" class, indicating a low false-alarm rate, which is critical for healthcare applications.

6. Conclusion

Securing the top position globally validates the effectiveness of the CNN-BiLSTM approach for Tamil. The model's ability to handle the specific nuances of Dravidian phonetics makes it a strong candidate for integration into large-scale, inclusive mental health diagnostic tools.